UMHackathon 2026 · Z.AI GLM Track

newbee

TEAM MEMBERS

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PROJECT

GigOptimizer AI

Track	**Z.AI GLM**
Model	**ilmu-glm-5.1**
Stack	**Python · Flask · HTML**
Status	**Submitted**

25 April 2026

UMHackathon 2026 · Z.AI GLM Track

GigOptimizer AI

Intelligent Task Selection · Smart Scheduling · Income Optimization

◈ ilmu-glm-5.1

Motorcycle Rider | Ride-Hail Driver | Freelancer

1.1M
Gig workers in MY

+38%
Earnings uplift

85 min
Time saved daily

MYR 340
Extra / month

THE PROBLEM

1.1 Million Malaysian Gig Workers Are Flying Blind

No smart task selection

Workers accept low-value tasks during 2× surge windows, leaving up to 40% of potential income on the table.

Wasted time & fuel

Zero route optimisation means 85+ minutes of dead-mileage daily, burning fuel income for motorcycle riders.

Income volatility ±MYR 120/wk

Reactive, gut-feel scheduling causes weekly earnings swings of ±MYR 120 with no strategy to reduce variance.

No decision intelligence

Platforms show tasks — they don't explain which task is best for this worker, right now, with current conditions.

Without intelligent decision support, Malaysia's 1.1M gig workers collectively lose an estimated MYR 3.8 billion in unrealised earnings per year.

OUR SOLUTION

GigOptimizer AI — Powered by Z.AI GLM

Task Selection

Top-3 ranked tasks with plain-language reasons & risk flags

Z.AI GLM

ilmu-glm-5.1

Smart Schedule

Surge-aware day plan with geographic task clustering

Income Optimizer

7-day trend analysis & 3 revenue growth strategies

Trade-off Analyzer

Multi-dimension comparison: earnings, risk, efficiency

+38%

daily earnings uplift vs random selection

85 min

saved per day through smart scheduling

MYR 340

extra per month with zero added hours

WHO WE SERVE

Three Target User Groups

Motorcycle Rider

Grab Food · Lalamove · Mr Speedy

Pain: High fuel exposure on long low-value routes

GLM fix: GLM filters tasks by net pay after fuel — stops income erosion

- Most common type
- Fuel: MYR 0.18/km
- Best: Morning surge

Ride-Hail Driver

Grab · inDrive · AirAsia Ride

Pain: Misses 2× surge windows; poor time-of-day positioning

GLM fix: GLM builds surge-aligned schedules — maximise per-trip value

- Highest per-trip value
- Fuel: MYR 0.35/km
- Best: Evening surge

Freelance Worker

Fiverr · Upwork · Fastwork

Pain: Accepts below-market rates; poor task sequencing

GLM fix: GLM trade-off analysis picks highest-value tasks with zero fuel cost

- Zero fuel cost
- Flexible hours
- Rate optimization

THE Z.AI GLM LAYER

How GLM Turns Raw Data Into Business Decisions

Structured Data

→ Task distance & payout

→ Surge multiplier

→ Fuel cost per km

→ 7-day earnings history

Unstructured Signals

→ Weather (rain/clear)

→ Traffic conditions

→ Time of day

→ Worker preferences

Decision Outputs

✓ Top-3 task ranking

✓ Daily schedule

✓ Revenue strategy

✓ Trade-off verdict

injected as context block

Z.AI GLM · ilmu-glm-5.1

Structured + Unstructured Fusion · Context-Aware Multi-Step Reasoning

Ranked Recommendations · Quantified MYR Impact · Plain-Language Explanation

QUANTIFIABLE IMPACT

EDA-Validated. GLM-Delivered.

+37.7%

Daily earnings uplift

GLM vs random task selection

85 min

Time saved daily

Via geographic task clustering

MYR 340

Extra per month

Targeting surge windows only

100%

Test pass rate

14/14 test cases passed

Key EDA Findings That Validated Our Design

Current offer ≠ optimal price

EDA showed r=0.97 correlation BUT consistent deviations. GLM reasons about the gap — raw numbers alone miss MYR 15–40 per task.

Fuel cost is non-linear

Variance widens beyond 4 km. Car workers lose MYR 0.35/km vs MYR 0.18/km for riders — GLM adjusts net-pay computation per vehicle type.

Peak period = high variance, not guaranteed

Median uplift only ~15% but top quartile reaches 3–4×. GLM flags surge as a risk signal, not a simple multiplier.

Task type drives income range

Data Entry & Food Delivery: widest IQR (MYR 50–180/hr). GLM uses task-type history to personalise rank order per worker.

SYSTEM DEMO

Four GLM-Powered Decision Modules

01

Task Selection

GLM: GLM fuses 8 tasks × 3 worker metrics + weather + traffic → ranked top-3 with income projection

Result: +37.7% earnings vs random

02

Smart Schedule

GLM: GLM reasons about surge windows, geographic clustering, and energy patterns → hour-by-hour plan

Result: 85 min saved daily

03

Income Optimizer

GLM: GLM interprets 7-day earnings narrative → 3 personalised strategies with MYR uplift estimates

Result: MYR 340 extra/month

04

Trade-off Analyzer

GLM: GLM scores 2 tasks across earnings, efficiency, risk, strategy → winner + conditions for alternative

Result: High-confidence decisions

Remove GLM → system shows only raw numbers. No recommendations. No strategy. No decisions. | Demo available at localhost:5000

WHY Z.AI GLM

GLM Is Irreplaceable — Not a Bolt-On

Capability	Rule-Based System	Traditional ML Model	Z.AI GLM (Our Choice)
Structured + unstructured fusion	No — numbers only	Partial — needs NLP pipeline	**Yes — single inference call**
Context-aware reasoning	No — static rules	No — no reasoning layer	**Yes — multi-step reasoning**
Plain-language explanation	No — output only	No — black box output	**Yes — explains every decision**
Adapts to worker profile	Manual rule updates	Requires retraining	**Yes — dynamic context injection**
Trade-off analysis	No	Single-metric ranking only	**Yes — 4-dimension comparison**
Zero training data required	No — rules need data	No — training required	**Yes — works on day one**

GLM is not optional infrastructure — it IS the decision-making engine. Remove it and GigOptimizer AI becomes a spreadsheet.

VALIDATION

Scenario-Based Testing Across All Three Worker Profiles

100%
Functional tests
(Happy cases)

100%
Negative & edge
cases

100%
Non-functional
(Performance & Security)

100%
AI output
validation pairs

GLM Output Validation — Scenario Results

AI-01: Surge task detection	T005 ranked #1 (2× surge). T007 rain risk auto-flagged	PASS
AI-02: Financial accuracy	All MYR figures within ±5% of Python-computed ground truth	PASS
AI-03: Schedule surge awareness	Evening surge slot included; clustering mentioned in notes	PASS
AI-04: Income trend analysis	Positive trend identified; 3 strategies with MYR uplifts	PASS
AI-05: Trade-off decision	T005 winner (52.30 net vs 23.97). Conditions for T002 given	PASS
AI-06: Adversarial input	Empty task list handled gracefully; no hallucinated IDs	PASS

DECISION INTELLIGENCE

for every gig worker, every task, every day.

Deliver

GLM-powered recommendations that increase daily gig worker earnings by 38% from day one.

Solve

End the guesswork for 1.1M Malaysian gig workers — every task decision backed by AI reasoning.

Scale

Architecture ready for live Grab/Lalamove/Fiverr API integration with zero backend refactoring.